Automated Interpretable Neuron Screening for Calcium Imaging

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***Abstract*—Calcium imaging records the activity of many individual neurons simultaneously by measuring activity-dependent fluorescence. Extracting clean per-neuron signals from the resulting video is a bottleneck: automatic component-extraction software is deliberately oversensitive, so a human expert must inspect hundreds to thousands of candidate cells per recording and reject the false ones—hours of subjective work per dataset. We automate this quality-control step with an interpretable machine-learning system. Redundant detections of the same neuron are first merged by a correlation-network step; each remaining candidate is then described by interpretable shape- and activity-based features, including wavelet descriptors of its calcium events, and classified by a glass-box model that reports the contribution of every feature to each decision. Validated by session-stratified cross-validation on a large hippocampal dataset (92,242 candidates from 187 recordings of freely-behaving mice), the system reproduces expert decisions (0.976 precision, 0.933 recall) in minutes rather than hours.**

***Index Terms*—Calcium imaging, quality control, machine learning, interpretable models, wavelet transforms, correlation networks**

I. Introduction

Understanding how neural populations encode behavior requires recording many neurons at once. Calcium imaging provides this: genetically encoded sensors produce fluorescence when a cell is active, and a miniature head-mounted microscope (a "miniscope") records the signal in a freely moving animal. From the resulting video, software must isolate each neuron's individual signal—a source-separation problem solved by algorithms such as CNMF-E [1] (in the CaImAn suite [2]). By design these favor sensitivity over specificity: rather than risk missing a real cell, they over-detect, so the output mixes genuine neurons with many spurious *candidates*—from movement, out-of-focus background fluorescence, optical artifacts, and a single cell being split into several copies (worsened by the limited depth resolution of one-photon imaging).

The standard workflow therefore ends with a manual step in which an expert inspects every candidate, weighing its shape in the image (its *spatial footprint*), its activity over time, and various quality metrics, and labels it a neuron or an artifact. For a typical session of 500–1500 candidates this requires one to three hours, and the implicit criteria that distinguish a "real" neuron are inconsistent between annotators and hard to transfer.

We automate this bottleneck with an interpretable pipeline whose contribution is fourfold: (i) a correlation-network stage that removes duplicate components by clustering co-active footprints; (ii) a 35-dimensional interpretable feature space, including calcium-event descriptors from wavelet analysis; (iii)

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a glass-box (fully inspectable) Explainable Boosting Machine (EBM) trained to penalize the costlier error type; and (iv) validation at scale on a large expert-labeled dataset, yielding expert-level accuracy with a complete decision audit trail.

II. Methods

A. Animals, imaging, and preprocessing

The system was developed on data from a large hippocampal imaging program [3]; we summarize the acquisition relevant to quality control. Adult C57BL/6 mice (with 5xFAD animals in a memory-impairment arm) were used. Neural activity was reported by the genetically encoded calcium indicator GCaMP6s—a slow, high-sensitivity sensor whose calcium transients last from a few hundred milliseconds to several seconds, which sets the time scale of the events analysed below. It was delivered by adeno-associated virus (AAV1.CAG.GCaMP6s, 300 nL) into dorsal hippocampal area CA1 (AP -1.94, ML $+1.46$, DV -1.2 mm) under anesthesia; one week later a 1-mm GRIN lens was implanted above CA1, and a baseplate for a miniature microscope (UCLA Miniscope V4.4) was mounted in a third procedure. After recovery, CA1 population activity was imaged in freely moving animals across four free-exploration and contextual-memory paradigms—among them cue-rich open-field exploration in a 44×44 cm arena over four daily 10-min sessions, contextual fear conditioning, a patterned open field, and a repeated open-field memory task. Overhead behavioral video was synchronized with the imaging stream in Bonsai [4]. All procedures were approved by the Bioethics Commission of Lomonosov Moscow State University (application 159-g, meeting 154-d-r, 17 August 2023).

Videos were motion-corrected with NoRMCorre [5] and decomposed into spatial footprints and calcium traces with CNMF-E [1] in the authors' BEARMIND pipeline (`github.com/iabs-neuro/bearmind`), built on CaImAn [2]. CNMF-E used estimated neuron size 3–6 px (at 3 μm/px), local-correlation threshold 0.85–0.95, and peak-to-noise ratio 3–15, with a first pass keeping components of spatial correlation $\geq$ 0.95 and SNR $\geq$ 3. Traces were normalized as dF/F (relative fluorescence change $\Delta F/F$) and footprints registered across sessions with CellReg [6]. These steps yield 92,242 candidate components from 187 sessions, each manually labeled as a genuine neuron (81.1%) or an artifact (18.9%)—the decisions our system learns to reproduce.

B. Network-based duplicate removal

CNMF-E frequently splits one physical neuron into several near-identical copies. Left uncorrected, such duplicates inflate population counts and inject artificial correlations, biasing downstream analyses of neural coding and population dimensionality and risking false conclusions. We remove this redundancy with a correlation-network step. Treating candidates as nodes, we connect two components when their dF/F traces are Spearman-correlated ($\rho \geq 0.4$) reproducibly across temporal blocks of the recording. The connected components of the resulting graph are groups of putative duplicates, ranked by their mean internal correlation (a graph-density score). Within each group, components whose footprints are spatially adjacent are taken to be the same cell: the highest-SNR copy is retained; a near-duplicate is merged into it when their SNRs are similar, otherwise the weaker copy is discarded. The spatial-adjacency requirement guards against collapsing genuinely distinct but co-active neurons, which are correlated yet spatially separated. In the deployed pipeline this deduplication precedes classification; the supervised evaluation reported here uses the full set of labeled candidates.

C. Interpretable feature space

Each surviving candidate is described by 35 features in six groups: *spatial/morphological* (area, circularity, convexity, aspect ratio, eccentricity, edge distance, local density, nearest-neighbour distance, ...), *temporal/trace* (skewness, kurtosis, baseline drift, noise, bimodality, Hurst exponent, ...), *event-based* (rate, fit quality, signal-to-noise, rise/decay times, amplitude variability, half-crossing rate), *reconstruction* (R^2, NMAE, NRMSE, residual SNR of a calcium-kinetics fit), *extractor-provided* (CaImAn SNR and spatial-correlation score), and a *kinetics-success flag* (see Sec. II-E). No single metric separates neurons from artifacts; the discriminative power is distributed across these complementary views.

D. Wavelet-based event characterization

The most discriminative features describe calcium transients, which we detect in the time–scale plane with a wavelet-ridge method [7] implemented in our DRIADA toolkit [8]. The dF/F trace is smoothed and mapped by a continuous wavelet transform using generalized Morse wavelets $\Psi_{\beta,\gamma}(\omega) \propto \omega^{\beta} e^{-\omega^{\gamma}}$ with $\beta = 2$ and $\gamma = 3$—an analytic, low-oscillation wavelet well matched to the asymmetric rise and decay of calcium events; scales span transient durations of 0.275–2.25 s, matching GCaMP6s kinetics. Local maxima of the transform are linked across scales into ridges, and a ridge is accepted as an event when it is sufficiently long, strong, and slow (thresholds on ridge length, peak magnitude, maximal scale, and duration). From the detected events E we derive descriptors such as event rate, amplitude variability, rise/decay times, an event signal-to-noise ratio, and the event-restricted reconstruction quality

$$R^2_{\mathrm{ev}} = 1 - \frac{\sum_{t \in E}(x_t - \hat{x}_t)^2}{\sum_{t \in E}(x_t - \bar{x}_E)^2}, \tag{1}$$

measured against a single-compartment kinetics fit $\hat{x}$ of the trace x.

E. Cascading kinetics estimation

The event features above are among the most discriminative, but they exist only when transient detection succeeds—and strict wavelet detection succeeds on just 42% of candidates, failing on cells with low signal-to-noise, sparse firing, or a drifting baseline. Discarding those cells would throw away exactly the borderline neurons a screening tool most needs to judge. We therefore characterize each candidate with the most reliable method that works for it, trying progressively more permissive options in turn: strict wavelet detection first; then wavelet detection with relaxed thresholds (≥ 3 events, $R^2_{\text{ev}} \geq 0.6$); then a robust amplitude-threshold detector, again strict then relaxed; and, when all else fails, physically plausible default kinetics ($t_{\text{rise}} = 0.25$ s, $t_{\text{off}} = 2.0$ s). This cascade raises the fraction of candidates with usable event features from 42% to 80%. Crucially, which tier succeeded is itself recorded as a feature, so the classifier knows how far to trust the event evidence for each cell; and because spatial, trace, and reconstruction features remain defined even when detection falls back to defaults, no candidate is ever left wholly uncharacterized.

F. *Classifier and decision objective*

Candidates are classified by an Explainable Boosting Machine (EBM) [9], [10], a generalized additive model with logit link g,

$$g\big(\mathbb{E}[y \mid \boldsymbol{x}]\big) = f_0 + \sum_j f_j(x_j) + \sum_{(j,k)\in\mathcal{I}} f_{jk}(x_j, x_k), \qquad (2)$$

with intercept f_0, per-feature shape functions f_j, and $|\mathcal{I}| = 20$ automatically selected pairwise interactions. Unlike black-box ensembles, every prediction decomposes exactly into additive feature contributions, yielding a complete audit trail; its accuracy is benchmarked against black-box baselines in Sec. III. Because errors are asymmetric—a false positive (artifact kept) can corrupt population statistics, whereas a false negative merely costs statistical power—we optimize

$$F_\beta = (1+\beta^2)\,\frac{\text{precision}\cdot\text{recall}}{\beta^2\,\text{precision} + \text{recall}}, \qquad (3)$$

with $\beta = \sqrt{1/3} \approx 0.577$ [11] (the F-measure weight, distinct from the wavelet exponent β of Sec. II-D), weighting false positives three times more than false negatives. Hyperparameters were chosen by stratified cross-validation over 128 configurations. The classifier was then refined over eight active-learning iterations—each adding expert corrections of borderline cases, 1129 in total—which raised cross-validated F_β from 0.955 to 0.965 and AUC from 0.961 to 0.978. At deployment, candidates may be filtered by expert-defined transparent rules (e.g. rejecting footprints below 3 μm^2, near-circular shapes, or candidates with no detected events), by EBM score at a tunable threshold, or by a hybrid that applies the rules first and the EBM to the remainder.

III. Results

A. *Classification performance*

We evaluate by cross-validation stratified *by recording session*, so that candidates from one session never appear in both training and test folds; this precludes exploiting session-specific imaging conditions. Across folds (mean $\pm$ s.d.) the production model attains $F_\beta = 0.965 \pm 0.004$, precision 0.976 ± 0.003, recall 0.933 ± 0.007, and AUC 0.978 ± 0.005. Consistent with the cost-sensitive objective, this operating point favours precision over recall. The low cross-session variance confirms stable generalization across paradigms. Because the same neurons recur across days within an animal, we also re-evaluated under a stricter *animal-stratified* split that shares no animal (hence no cross-day-tracked neuron) between train and test: a random-forest probe showed no degradation (F_β $0.948 \to 0.959$, AUC $0.960 \to 0.965$), so the reported accuracy reflects cross-animal generalization rather than memorization of recurring neurons.

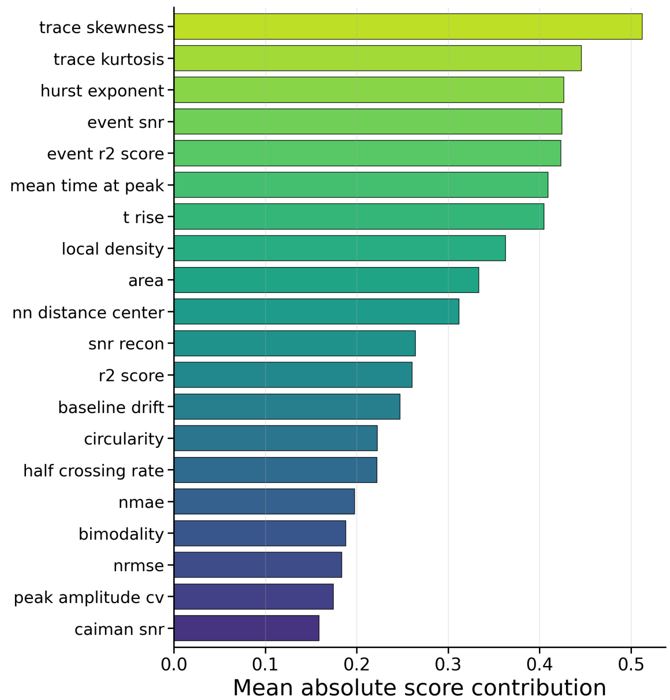

Fig. 1. Mean absolute score contribution of the top features in the production EBM. Temporal trace statistics (skewness, kurtosis) and the Hurst exponent dominate, followed by wavelet event descriptors (event SNR, event R^2); the extractor-provided CaImAn SNR ranks lowest.

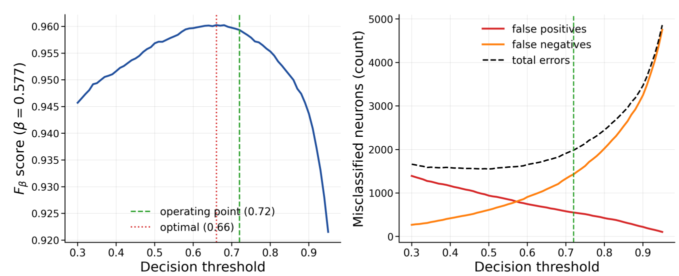

Fig. 2. Threshold analysis on the held-out test split. Left: F_β (β=0.577) versus decision threshold, with the deployed operating point (0.72) and the F_β optimum (0.66) marked. Right: false positives, false negatives, and total errors versus threshold.

Table I benchmarks classifiers on a single held-out split, so its figures fall slightly below the cross-validated headline above. CaImAn's own spatial-correlation and SNR metrics are weak discriminators on one-photon data (AUC 0.59 and 0.56 individually); even an optimally tuned threshold on both

TABLE I
TEST-SPLIT COMPARISON (THRESHOLD TUNED ON TRAINING DATA; $\beta = 0.577$).

Method	Prec.	Rec.	F_β	AUC
CaImAn metrics (best rule)	0.836	0.974	0.867	—
Logistic regression	0.965	0.912	0.951	0.946
Random forest	0.939	**0.975**	0.948	0.960
EBM (ours)	**0.968**	0.938	**0.960**	**0.966**

(Table I, top row) attains only $F_\beta = 0.87$—the value already obtained by accepting every candidate at the dataset's 81% positive rate—so the CaImAn metrics carry little discriminative information, motivating multi-feature screening. Among learned classifiers (Table I, held-out test split), off-the-shelf logistic regression and random forests reach $F_\beta \approx 0.95$, so the task is learnable; the glass-box EBM matches or exceeds both ($F_\beta = 0.96$, AUC 0.97, highest precision) while remaining fully interpretable—interpretability here costs no accuracy.

B. *Which features matter*

Feature-importance analysis (Fig. 1) shows that temporal trace statistics carry the strongest discriminative power: trace skewness and kurtosis rank first and second, validating the heavy-tailed, "spiky" signature of genuine calcium dynamics. Wavelet-derived event descriptors rank immediately below, whereas the extractor's own amplitude SNR ranks lowest—temporal structure is far more informative than raw amplitude.

C. *Tuning the operating point*

Because the EBM emits a calibrated score, a single threshold tunes the precision/recall balance (Fig. 2). The unconstrained F_β optimum lies near a threshold of 0.66; the deployed operating point (0.72) sits slightly above it, trading a little recall for additional precision, consistent with the asymmetric-cost objective. Lower thresholds (0.3–0.4) maximize recall for exploratory screening.

D. *Practical impact*

Manual curation of a single session takes one to three hours. With the system presented here, the same session—duplicate removal, feature extraction, and classification—is processed in two to three minutes on a standard workstation, almost two orders of magnitude faster, enabling same-day analysis. Identical criteria applied to every candidate make the output reproducible, and the additive decomposition of Eq. 2 shows exactly why each decision was made.

IV. DISCUSSION AND CONCLUSION

We have presented an automated quality-control system that automates the principal manual bottleneck in calcium-imaging preprocessing. Removing duplicates by clustering a correlation network (Sec. II-B) and characterizing activity through wavelet-detected events (Sec. II-D) supply the features that prove most discriminative. Interpretability also enables iterative improvement: the additive structure of Eq. 2 exposes which features drove a decision, so a corrected label becomes a training example for the next round.

Limitations remain: the system was trained on one-photon GCaMP recordings of hippocampal neurons, so transfer to other modalities or cell types may require retraining; entirely silent neurons are inherently ambiguous; and $\sim$20% of candidates still rely on default kinetics. Natural extensions include tighter coupling to upstream extractors, real-time assessment during acquisition, and session-specific calibration. The broader point is practical: an interpretable model here reaches the accuracy of black-box classifiers while exposing the reason for every decision, and, combined with CNMF-E extraction, reduces curation to reviewing a small set of flagged borderline cases.

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